Data Discovery Report — MYH7 knockout vs wild-type cardiomyocyte gene regulatory networks

> ⚠️ **Synthetic example.** This report was generated by running the real pipeline against the mock collection built by `scripts/make_mock_inventory.py`. Every figure below is a genuine output of that run, but the underlying files, request IDs and paths are fabricated. It exists to show the format; it describes no real dataset.

Requested by: example user · **Date:** 2026-11-04 · **Run:** `run-mock-example`

Goal as understood:

> *Find existing data that could support a comparison of gene regulatory networks between MYH7-knockout and wild-type cardiomyocytes, preferring processed matrices over raw reads.*

Scope searched: collection `MOCK Example Lab Collection` · roots `/lab_data/`, `/rigs/`, `/core_imaging/` · types `all`
Corpus: 169 files indexed · 133 matched by CSV · 27 inferred · 9 unattributed

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1. Recommended datasets

Ordered by fit. Grouped by sequencing request, not by file.

Unattributed — multiome reanalysis directory

Fit: The only material in the collection that names both genotypes and a GRN output. Directly on target, but carries **no request record** — see the caveat below.
Project: — · **Requester:** unknown · **Status:** not in the CSV
Platform: not recorded
Original goal on file: none

PATH	ROLE	SIZE
`/lab_data/reanalysis/myh7_ko_vs_wt_cardiomyocyte_multiome/grn_edges.tsv`	inferred GRN edge table	334 MB
`/lab_data/reanalysis/myh7_ko_vs_wt_cardiomyocyte_multiome/atac_peaks.bed`	inferred ATAC peaks	544 MB

> ⚠️ This directory reached the `inferred` tier at confidence 0.75. Its meaning comes from folder naming, not from a request record. Nothing verifies that these edges were computed from MYH7-KO and wild-type cardiomyocytes, or which reference build was used. Confirm with whoever generated it before building on it.

REQ-1091, REQ-1238 — 10x single-cell RNA-seq

Fit: Single-cell expression with filtered matrices ready to load. Neither request names a genotype, so they can serve as a reference atlas, not as the comparison itself.
Project: Cardiac regeneration · **Requester:** Jones, Lee / Garcia, Max · **Status:** Completed
Platform: HiSeq / NextSeq 2000
Original goal on file: not recorded

PATH	ROLE	SIZE
`/lab_data/sequencing/abc/REQ-1091/H60QDRX3/outs/filtered_feature_bc_matrix.h5`	filtered count matrix	200 MB
`/lab_data/sequencing/def/REQ-1238/H42MDMX2/outs/filtered_feature_bc_matrix.h5`	filtered count matrix	600 MB

Shown 2 of 14 files. Full listing in `knowledge/ASSOCIATIONS.md`.

REQ-1000, REQ-1119 — bulk RNA-Seq

Fit: Bulk expression from the same lab, usable as a low-resolution sanity check on any GRN-derived expression claim. Not single-cell, so it cannot support network inference on its own.
Platform: HiSeq / Unknown

PATH	ROLE	SIZE
`/lab_data/sequencing/abc/REQ-1000/H51EDRX3/counts/gene_counts.tsv`	count matrix	500 MB
`/lab_data/sequencing/jkl/REQ-1119/H81ADMX2/counts/gene_counts.tsv`	count matrix	600 MB

2. Possibly relevant — inferred metadata only

These matched on folder naming, not on a request record. **Descriptions are inferred and unverified — confirm before relying on them.**

PATH	INFERRED AS	CONFIDENCE
`/lab_data/reanalysis/myh7_ko_vs_wt_cardiomyocyte_multiome/`	MYH7 knockout vs wild type, cardiomyocytes, 10x Multiome	0.75
`/rigs/patch_clamp/cm_myh7_ko/`	patch-clamp, cardiomyocytes, MYH7 knockout	0.75
`/lab_data/reanalysis/sc_LV_d14_mi/`	single-cell, left ventricle, myocardial infarction, day 14	0.65

3. What these data cannot answer

3a. Unanswerable with any data in this collection

- **Differential GRN structure between genotypes at defined timepoints.** The one multiome directory records no timepoint, no replicate structure, and no genotype assignment per sample. *Requires:* the sample sheet for that reanalysis, or a re-run from the source matrices with genotypes declared.
- **A physiological readout for both genotypes.** `/rigs/patch_clamp/cm_myh7_ko/` suggests patch-clamp recordings from knockout cardiomyocytes only; no wild-type recordings are named anywhere. *Requires:* matched wild-type recordings, or confirmation that the folder holds both.
- **Statistical power for edge-level comparison.** Replicate count is unrecoverable from filenames alone. *Requires:* the original request record.

3b. Requested but never delivered

- **6 of 40 requests** have no result path recorded in the metadata CSV: REQ-1056, 1063, 1112, 1161, 1189, 1259. REQ-1161 is a 10x Multiome request (status Canceled), the one assay type that would have mattered here.

3c. Indexed but absent from the scanned scope

- **11 of 40 requests** have a CSV path but no matching files under `/lab_data/` : REQ-1007, 1028, 1084, 1126, 1140, 1168, 1196, 1231, 1245, 1266, 1273. These may be untransferred or outside the scanned roots. Worth checking before concluding the data does not exist.

3d. Present but undocumented

- **9 files (2.8 GB)** carry no attributable metadata: no request anchor, and folder names too vague to interpret. Largest clusters: `/lab_data/new_data/final2/` (3 files), `/lab_data/tmp/backup/v3/` (2 files), `/lab_data/scratch/` (2 files). These may contain relevant data; nothing can currently say so.

Standing limitations

- `Analysis Result Paths` is unpopulated for all 40 requests, so no derived analysis output can be matched by request ID. Processed results, including the multiome directory above, reach the `inferred` tier at best.
- The metadata CSV indexes sequencing requests only. Proteomics, imaging, EM, behavior and ephys have no request number and cannot appear in it. Absence from the CSV is not absence from the collection.
- No `.zip` archives in scope. Had there been any, their contents would be invisible to this search until listed (workflow Step 3a).
- All requests are from one lab; no cross-lab comparison is supported.

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4. Suggested next steps

- Identify the author of `/lab_data/reanalysis/myh7_ko_vs_wt_cardiomyocyte_multiome/` and recover its sample sheet. This single artifact decides whether the question is answerable today.

- Check whether the 11 requests in §3c were transferred, before commissioning anything new.
- Add a README to `/lab_data/new_data/`. Its files are currently unattributable.

Generated by the Globus discovery agent. Associations recorded in `knowledge/file_associations.jsonl`; this run logged in `knowledge/query_log.jsonl`. Corrections welcome — they become permanent overrides.